

Syllabus

Fall semester 2025

Course Title	DNA science		Course Code	CB4205		Credit	3	Classroom	Hyung 302 Tam 501
			Type			Hours	3		
Teacher	Name	Class no.	Office	Office hours		Contact		E-mail	
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1. Course Description

Aims and objectives	Understand DNA by our brain and learn DNA by our hands doing experiments.
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2. Core Target Competencies

Subject Core Competencies				
1. Knowledge Comprehension	2. Critical Thinking	3. Problem-solving	4. Communication	5. Collaboration skills
1. Essential principles of genetics, and DNA structure and function (chap 1 to 3) 2. Small and large-scale methods for analyzing DNA(chap 4 to 6) 3. Experiments related with DNA (lab 1 to 12)				

3. Textbooks and References

A. text book :

DNA Science : A first course, 2nd ed

–David A. Micklos, Greg A. Freyer

B. reference book :

Essentials of Molecular biology, 4th edition

–George M. Malacinski

Watson et al, Molecular biology of the Gene. 5th Edition (International edition), Pearson (Benjamin Cummings).

R.F. Weaver, Molecular Biology. McGrawHill.

4. Class Method

Lecture / Experiment / Group presentation and discussion

5. Evaluation

A. Evaluation Criteria

Team Presentation ¹	Homework ²	Midterm exam ³	Final exam ³	Attendance and attitude ⁴	Total
10%	20%	30%	30%	10%	100%

B. Evaluation Method

1. There would be team presentation irregularly. You will belong to a team and will do group activity for presentation and will do experiments as a team member.
2. Prepare notebook for solving problems in "Result and discussion" and "For further research" in Laboratories section.
3. Exams will contain short answer questions and essay questions.
4. Absence without notice: -1pt, being late (every twice : -0.5pt)
5. The recognized score for an examination missed due to an official absence or medical leave shall be calculated as follows: the percentile score in the missed examination corresponding to the student's total percentile score obtained in other examinations of the same subject) $\times$ (100% for official absence; 80% for medical leave).

6. Criteria for Awarding Credits

☞ Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.

7. Weekly Plans

Week (period)	Topic and Contents	Class activity
1 (8/18~)	Lesson plan and orientation Chap 1. How we learned that DNA is the genetic material	Survey a scientist I
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
2 (8/25~)	Lab 1. Measurements, micropipetting, and sterile technique Lab 2. Bacterial culture technique	Results and Discussion/ For Futher Research
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
3 (9/1~)	Chap 2. How we learned the function of DNA	Survey a scientist II
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
4 (9/8~)	Lab 3. DNA restriction analysis Lab 4. Effect of DNA methylation on restriction	mapping of pX
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
5 (9/15~)	Chap 3. How we learned how genes are regulated	IL-4 and hemolysin gene analysis
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
6 (9/22~)	Lab 5. Rapid colony transformation of E. coli with plasmid DNA Lab 6. Assay for an antibiotic resistance enzyme	Results and Discussion/ For Futher Research
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
7 (9/29~)	Chap 3. How we learned how genes are regulated	Why is the stone formed in bladder and how is it cured?
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
Week 8 10.6.~10.10.	Chuseok Holiday, Hangeul Day, Discretionary Holiday	
9 (10/13~)	Midterm exam (10/13~10/17)	

Week (period)	Topic and Contents	Class activity
10 (10/20~)	Chap 4. Basic tools and techniques of DNA science	Drawing presentation Presentation on model organisms
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
11 (10/27~)	Lab 7. Purification and identification of recombinant GFP (part A) Lab 8. Purification and identification of plasmid DNA	Results and Discussion/ For Futher Research
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
12 (11/3~)	Chap 5. Methods for finding and expressing important genes	Drawing presentation Presentation on MB techniques
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
13 (11/10~)	Lab 9. Recombination of antibiotic resistance gene Lab 10. Transformation of E. coli with recombinant DNA	Results and Discussion/ For Futher Research
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
14 (11/17~)	Chap 6. Modern methods for analyzing whole genomes	Study on synthetic genome
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
15 (11/24~)	Lab 11. Replica plating to identify mixed E. coli population Lab 12. Purification and identification of recombinant DNA	Results and Discussion/ For Futher Research
Core Competencies	3. Problem solving/4. Communication/5. Collaboration skill	
16 (12/1~)	Chap 6. Modern methods for analyzing whole genomes	
Core Competencies	1. Knowledge comprehension / 2. Critical thinking	
17 (12/8~)	Final Exams	

※ The above plans may change according to the academic schedule.